

This is to build up some intuition about option pricing and trading strategies. We will use the following notation:

- S_0 : Current stock price
- K : Strike price of option
- T : Time to expiration of option
- S_T : Stock price at maturity
- r : Continuously compounded risk-free rate of interest for an investment maturing in time T
- C : Value of American call option to buy one share
- P : Value of American put option to sell one share
- c : Value of European call option to buy one share
- p : Value of European put option to sell one share
- D : Present value of dividends paid during the life time of the option

1 Interesting Portfolios

Consider the following 4 portfolios for a non-dividend-paying stock

- Portfolio A: one European call option plus a amount of cash equal to Ke^{-rT}
- Portfolio B: one European put option plus one share
- Portfolio C: one share
- Portfolio D: an amount of cash equal to Ke^{-rt}

These 4 portfolios should be very reasonable and intuitive to think about. Portfolio A holds a call with potential cash to buy stock. Portfolio B holds a put with a share to potentially sell. Portfolio C holds a single share while portfolio D holds just cash. At time T , these 4 portfolios are:

- Portfolio A: if $S_T \leq K$, option worths nothing, the cash equals to K . if $S_T > K$, we exercise the option by using the cash that is equal to K to to buy one share which is valued at S_T . So the value of the portfolio is $\max(S_T, K)$
- Portfolio B: if $S_T \geq K$, option worths nothing, we have one share that is valued at S_T . if $S_T < K$, we exercise the option by selling the share at K and we hold cash equal to K at the end. So the value of the portfolio is $\max(S_T, K)$.
- Portfolio C: one share

- Portfolio D: an amount of cash equal to K

This can be summarized as the following table content:

Portfolio	t=0	t=T	Note
A	$c + Ke^{-rT}$	$\max(S_T, K)$	long call with cash
B	$p + S_0$	$\max(S_T, K)$	protected put
C	S_0	S_T	underlying stock
D	Ke^{-rT}	K	cash

In scenarios the stock pays future dividend, denote D as the present value of the dividends during the life of the option. We would have the similar 4 portfolios:

- Portfolio A': one European call option plus a mount of cash equal to $D + Ke^{-rT}$
- Portfolio B': one European put option plus one share
- Portfolio C': one share
- Portfolio D': an amount of cash equal to $D + Ke^{-rt}$

2 Basic Boundaries

- American options are more valuable than European options: $C \geq c, P \geq p$
- call options can't be worth more than the stock: $C \leq S_0, c \leq S_0$
- put options gives the right to sell one share for K , so it can't be worth more than K : $p \leq K, P \leq K$

2.1 Boundaries for European call options

Consider Portfolio A and C. At time T , you can observe that the value of Portfolio A always worth as much as, and can be worth more than Portfolio C. *It follows that in the absence of arbitrage opportunities this must also be true today.* Hence,

$$c + Ke^{-rT} \geq S_0$$

Combine with the fact that option price needs to be positive and the boundary established above, we have:

$$\max(S_0 - Ke^{-rT}, 0) \leq c \leq S_0$$

Similarly, for a stock that pays dividends, we have:

$$\max(S_0 - D - Ke^{-rT}, 0) \leq c$$

2.2 Boundaries for European put options

Consider Portfolio B and D. At time T , you can observe that the value of Portfolio B always worth as much as, and can be worth more than Portfolio D. *It follows that in the absense of arbitrage oportunites this must also be true today.* Hence,

$$p + S_0 \geq Ke^{-rT}$$

Combine with the fact that option price needs to be positive and the boundary established earlier, we have:

$$\max(Ke^{-rT} - S_0, 0) \leq p \leq Ke^{-rT}$$

Similarly, for a stock that pays dividends, we have:

$$\max(D + Ke^{-rT} - S_0, 0) \leq p$$

3 Pull-Call Partity

3.1 European options

Consider Portfolio A and B. At time T they always worth the same. Because they can't be exercised before the expiration at time T , the portfolios must therefore have identical values today. Hence

$$c + Ke^{-rT} = p + S_0$$

Similarly, for a stock that pays dividends, we have:

$$c + D + Ke^{-rT} = p + S_0$$

3.2 American options

3.3 Trading Strategies with a single option

Let's consider trading strategies that can be explained by put-call parity.

$$p + S_0 = c + Ke^{-rT} + D$$

This can be read as: a long position in a put with a long position in the stock is equivalent to a long call position plus a certain amount of cash $Ke^{-rT} + D$. This is a **protected put**.

Similarly consider this form of put-call parity:

$$S_0 - c = Ke^{-rT} + D - p$$

This can be read as: a short position in a call with a long position in the stock is equivalent to a short put position plus a certain amount of cash $Ke^{-rT} + D$. This is a **covered call**.